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Genetic manipulation for enhanced aquaculture production in Nigeria

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ABSTRACT: Aquaculture in Nigeria has witnessed some expansion in recent times that this is fairly insignificant relative to world standard. With the demand for fish increasing everyday, there is the need for new approach towards increasing fish production. The application of genetics can play a significant role in efforts to increase aquaculture production. This paper then review the importance of genetics application in fish, various methods of genetic manipulation such as selective breeding, hybridization, transgenics ploidy manipulation and molecular marker assisted selection were thoroughly discussed. Also various constraints limiting the application of these methods in Nigeria, such as insufficient funding, limited research facilities lack of appropriate skills and technical know how, little or no awareness about the potential of fish genetics. Inadequate aquaculture extension services and hostility from the public were explicitly elucidated. Also, solutions and way forward to these constraints were proffered. Applications of these techniques in conjunction with local methods will hopefully contribute to the development of more efficient aquaculture.

Keywords: Genetic manipulation, Fish, Aquaculture, Breeding, Production

Introduction

Fish is an important source of food and income to many people in developing countries. In many parts of the world several people depend wholly or partly on the fisheries sector for their food and livelihood (FAO 2008). The demand for fish and fishery product has increased tremendously all over the world, due to population explosion and increased nutritional benefits over meat from livestock (FAO, 1997 Gupta and Acosta, 2004). Supply of fish from the wild have been over exploited in recent years and have reached maximum sustainable yield as a result of pollution of water bodies, over fishing and environmental degradation (Horemans, 1998). Aquaculture will need to be developed progressively. According to Jamu and Ayinla (2003) the future of aquaculture in the country, depends on increasing production efficiencies and intensities by using less land, water, feed and financial investments.

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While there is opportunity for increasing production from aquaculture venture through improved farm management practices, quality feed and good water quality, the increases in production needed to meet the demand for fish globally will have to come from the use of genetically improved breeds/strains, as has been the case in crops and livestock (Gupta and Acosta, 2001). In most of the developing countries across the world, the use of genetic research findings and breeding programmes has made tremendous impact in crop production and it has revolutionize the entire livestock industry in recent years (FAO, 2001). Fish genetic manipulation is a sure means of enhancing yield in aquaculture to meet the ever-increasing but unmet need in animal protein. The sustainable utilization of genetic resources including fish is a vital part in improving the standard of living in a populated country like Nigeria. Concern over declining harvest and a obvious reduction in abundance and bio-diversity of fish species from the wild has led to a more holistic approach in fisheries management and research in fish genetics so as to enhance food security and sustainable development in aquaculture (Wong et al., 2008).

Teleosts, unlike commercially important higher vertebrates have certain reproductional and biological properties (e.g. external fertilization and embryogenesis) allowing easy manipulation of genetic and physiological processes in the early stages of ontogenesis, for producing altered phenotype (Horvath and Orban, 1995). To sustain the rapid growth and expansion in aquaculture, fish genetics are being exploited for their potential to increase aquaculture production, efficiency and yields as a mean of improving protein in our diet, for food security and poverty reduction. According to Akinrotimi *et al.* (2007) effective strategy for reducing poverty involves creation of income earning opportunities for the people by increasing productivity through improved technology.

This paper then reviews critically the limitation of species used in aquaculture in enhancing yield, the importance of and methods of genetic manipulation in fish for improved yield, and strategies in aquaculture, constraints and strategies for the applications of genetic manipulation to boost fish production, enhance food security and improve livelihoods of the ever increasing population of Nigeria riddled with poverty.

Species used in aquaculture

For any species of fish to be considered as aquaculture candidate it must possess culturable qualities such as hardness, fast growth rate, easy adaptation to unfavourable environmental conditions, ability to convert feed to carcass weight and wide acceptability. This depends largely on the production function. Farmers references and choice of final consumers (Akinrotimi et al., 2009). Since of the culturable species utilized in aquaculture are in the early stages of domestication their response to new and enhanced management practices is limited (Jamu and Ayinla, 2003). Hence, for any noticeable and appreciable production gains emphasis should be placed on developing high performing and result oriented strains through genetic manipulations (Eknath *et al.* 1993). In Nigeria, Nile tilapia *Oreochromis niloticus* is the most commonly culture fish, followed closely by African mud catfish *Clarias gariepinus* among others (Table 1).

However, these species are limited in their culture potential as a result of poor brood stock management, loss of genetic diversity in culture system, contamination of the wild and indigenous general and lack of baseline genetic data (Pullin and Capilli, 1988). To overcome these challenges and improve the performance of culture fish genetic manipulation of cultured species to allow faster growth and efficient feed conversion ratio is needed.

Table 1. Some Important Culturable Species in Nigeria

Species	Countries	Potential Yield
<i>Clarias gariepinus</i>	Very good	Very high
<i>Heterobranchus bidorsalis</i>	Very good	Very high
<i>Heterobranchus longifillius</i>	Very good	Very high
<i>Lates niloticus</i>	Very good	Very high
<i>Chrysichthys nigrodigitatus</i>	Very good	Very high
<i>Oreochromis niloticus</i>	Good	High
<i>Tilapia guineensis</i>	Good	High
<i>Heterotis niloticus</i>	Good	High
<i>Gymnarchus niloticus</i>	Good	High
<i>Parachanna obscura</i>	Average	Average
<i>Megalops atlanticus</i>	Low	Low
<i>Elops lacerta</i>	Low	low

Source: (Ugwumba and Ugwumba 2003).

Importance of Genetics in Aquaculture

The introduction of genetic techniques has the potential to increase significantly aquaculture production.

1. Enhanced feed utilization and growth

The possibility of enhancing growth rate through the introduction of additional growth hormones has been applied successfully in fish (Duelin *et al.*, 1994). Spectacular growth enhancement programmes has been demonstrated in some species which include salmonids. These studies have shown enhancement of growth in adult salmon to an average of 3-5 times, the size of non-transgenic controls, with some individuals, especially during the first few months of growth reaching as much as 10-30times the size of controls (Mori and Devlin, 1999). Dey *et al.* (2002), reported that the genetically improved farm tilapia, (GIFT) *O. niloticus* recorded 3times more than the non genetic improved tilapia (Table 2).

Table 2: Yield and variable costs tilapia farming using GIFT and non-GIFT strains in cages and earthen ponds in on-farm trails

Items	Bangladesh	China		Philippines		Thailand	
GIFT	Pond	Cage	Pond	Cage	Pond	Cage	Pond
Yield (kg/unit area)	1.593	3.893	4.645	236	1.361	2,829	743
Variable cost	463	4.191	3.548	168	1.385	1.510	427
Variable cost of fish/kg	0.29	1.08	0.76	0.71	1.02	0.53	0.58
Non GIFT							
Yield (kg/unit area)	896	3.111	4.275	153	912	2.044	558
Variable cost	405	4.191	1.523	168	1.375	1.517	411
Variable cost of fish/kg	0.45	1.35	0.82	1.1	1.51	0.74	0.74
Percentage difference (GIFT – non-GIFT/Non-GIFT)							
Yield (kg/unit area)	77.8	25.1	8.7	54.2	49.2	38.4	33.2
Variable cost	14.3	0	0.7	0	0.7	-0.3	3.9
Variable cost of fish/kg	-35.6	-20	-7.3	-35.5	-35.5	-28.4	-21.6

Source: Day 2002

In aquaculture, determines the profitability of the whole enterprise. It accounts for 65% of total cost (Gabriel *et al.*, 2007). Genetically manipulated species have improved efficient feed conversion into carcass gain. The role of nutrition in fish is to reduce the cost of production and increase profit (Dudley-Cash, 1996). Genetics provides the performance potential and the nutritionists exploits that performance to achieve maximum profit (Hardy, 1999).

3. Control of reproductive activity and improved resistance to disease

Conventional induction of spawning in fish largely depends on external stimulation through administration of hormones, natural or synthetic (Peter and Yu, 1997; Melamed *et al.*, 2002). On the other hand, reproductive activities in fish like tilapia needs to be controlled because of early sexual maturation and stunted growth. Culture of monosex progeny, preferably males, which in tilapia species with faster growth and larger size than females has long been recognized as the most effective solution (Mair *et al.*, 1997). This can be achieved by a number of methods such as exposure of fry to methyltestosterone at their early stage to produce all male tilapia. Also crossing of some

species with varying genetic make up (*Oreochromis niloticus* and *O. mossambicus*) can also results in hybrid of single sex (Melamed *et al.*, 2002).

Genetics has contributed immensely to the new approach of combating viral and bacterial pathogen in fish culture through the use of DNA vaccines and antimicrobial agents (Corbeil *et al.*, 2000), a major advancement in fish farming over the last decade. The use of the vaccines has reduced losses to diseases and increased the capacity of the fish for growth by reducing metabolic costs associated with fighting pathogens (Torrissen *et al.*, 1995). The application of vaccines in fish is based on the injection of naked DNA encoding part of the antigen so that the protein will be expressed *in vivo* and the production of antibodies induced. In Atlantic salmon, as other species this is done by injection of plasmid encoding infectious hematopoietic necrovirus (IHNV) glycoprotein under the control of the cytomegalovirus promoter (PCMV) (Traxler, *et al.*, 1999). Subsequent challenge with virus eight weeks later indicated that a significant measure of protection had been given through DNA vaccination. Similar protection was observed in rainbow trout after vaccination against viral haemorrhagic septicaemia virus (VHS), using the glycoprotein encoding sequence driven by PCMV (Lorenzen *et al.*, 1999).

METHODS OF GENETIC MANIPULATION IN FISH

Several genetic manipulation methods have been used to improve the performance of fish in aquaculture for decades; the application approach depends on the species of fish, age, sex, size, production function, environmental factors, level of technology, finance and availability of materials and resources. These techniques have been very successful in other countries and provide an ample opportunity for increasing aquaculture production in Nigeria.

1. Hybridization

The methods of hybridization is to combine useful, positive or desirable traits from two species with the aim of producing offspring of superior quality than the parents (Ayinla and Nwadukwe, 2003). Results obtained from hybridization process include the production of sterile progeny and sometimes monosex individuals as in tilapia. These are generally characterized by lower fecundities, larger egg size and consequently fast growing individuals (Nwadukwe, 1992). However, Brink *et al.*, (2002) observed that ecological risks have been reported from hybridization activities in some countries.

In the tilapia hybridization, Agnese *et al.* (1998) reported that crossing of female *Tilapia tholoni* with a male *O. mossambicus* yielded 100 percent females but crossing female *O. spirilus* with male *O. leucostictus* yield 98 percent males. While in catfish culture, Anyinla & Nwadukwe (2003) and Nwadukwe (1995) reported the hybridization process of crossing male *Heterobranchius bidorsalis* with *Clarias gariepinus* so as to improve the commercial value by fast growth (Table 3). Also, Aluko (1985) or 1995 reported the hybridization of *Claria anguillaris* and *H. longifilis* with the same purpose.

Table 3: Growth rate of Heteroclarias and *Clarias gariepinus*

Species	Growth Range	Rate (kg) mean
<i>Clarias gariepinus</i>	0.4 – 1.2	0.8
<i>Heterobranchius bidorsalis</i>	0.3 – 0.7	0.4
<i>Hetero clarias</i> (Hybrid)	0.6 – 1.5	1.0

Source: Ayinla and Nwadukwe, 2003

Legendre *et al.* (1992) in Ivory coast also reported the hybridization of *C. gariepinus* and *H. longifilis*. Their findings were comparable to the one obtained in Nigeria. In choosing hybridization method as a mean of enhancing the performance of the indigenous population in aquaculture, there should be well established genetic characterization records in order to monitor the long term purity of the parental lives (Chongadeya *et al.*, 2003).

2. Ploidy manipulations

A fish specimen carrying set(s) of chromosomes is called polyploidy (Thorgaard, 1986). The most important and applied ploidy manipulations in aquaculture are induced triploidy and tetraploidy Donaldson and Benfey, 1987;

Ihsse et al., 1990; Thorgaard et al., 1992). Triploidy is the condition of possessing three times the haploid number of chromosomes in the cell nucleus. A triploid animal is normally sterile because of the inability of homologous chromosomes to synapsis in meiosis (Gui et al., 1996). This involves the process of obtaining a sterile fish, that do not invest energy into reproduction activities but rather divert it into somatic growth (Abu et al., 2009). Ploidy manipulation results from the chromosomally imbalanced eggs or sperm for the process of egg and sperm formation through the embryo development and hatching to proceed successfully. The genetic code of the eggs and sperm must be perfect (Rottman et al. 1991). The cells of triploid individuals have three sets of chromosome (3n) as compared to the normal two sets, 2n (Cassani et al., 1990).

Triploids have been produced by preventing the second meiotic division after the sperm enters the egg, thereby resulting in two sets of chromosomes from the female and one set from the male (Cherfas et al., 1993). This procedure is usually accomplished through chemical method, heat shock, cold shock, electric shock and hydrostatic pressure.

Induction of triploidy has been reported in many fish species. Rainbow trout (Thorgaard, 1992), Atlantic salmon (Benfey and Sutterlin, 1984), Common carp (Chao et al., 1986), *Clarias gariepinus* (Abu et al., 2009), Pacific Oyster (Guo et al., 1996).

3. Tetraploidy and Selective breeding

Tetraploidy is a condition where a fish possesses four times (4n) the number of chromosomes in the nucleus of a cell. According to Harvath and Orban (1994), no tetraploid fish (4n) have been found so far in natural water, however duplication of the diploid genome might have occurred during phylogenesis, leading to the formation of new species. The same methods used in inducing triploidy is applicable in tetraploidy except that the timing of the shock is different. As opposed to sterile triploid individuals, tetraploid fish is fertile and suitable for production of triploids by crosses with diploid individuals (Cherfas et al., 1995, Guo et al., 1996).

Effective selective breeding programs helps in better utilization of feed, water and ultimately increase production (Hayes et al., 2006). The first step in starting selective breeding (Huang and Liao, 1990). This base population must capture enough genetic variation to ensure maximum long term genetic response from the breeding program (Gjedem et al., 1991). One strategy would be to measure all the traits in the breeding goal in the wild fish and select a base population which maximizes the variance of these traits (Toro et al., 1999).

Heritability estimates have been made for traits important to aquaculture in fish and selection programmes have yielded impressive gains in fish performance based upon these heritability estimates (Gjedem et al., 1991; Su et al., 1996). According to Gupta and Acosta (2004) before initiating a selection program in fish it is very important to first of all identify the breeding goal and the traits of economic importance, such as disease resistance, growth response, meal quality among others. Traits may be positively or negatively genetically correlated, as a primary responds to a selection. Secondary traits will respond depending on the nature of genetic correlation between the primary and secondary traits (Anon, 1988).

Selective breeding have been employed successfully especially in tilapia culture leading to higher yields (Fitzsimmons, 2000). Popular Nile tilapia with good traits were carefully selected from some countries across the globe (Table 2). To understand their morphometric and genetic differences, the assembled *O. niloticus* were characterized using geographic, morphometric, electrophoretic and molecular biometer techniques (Eknath et al., 1991). The selectively breed strain was named GIFT fish (Genetically improved farmed tilapia). The GIFT project ended in 1997 after completing six generations of selection. The genetic gain per generation was about 17 percent across five generations of selection for growth (Eknath and Acosta, 1998).

Molecular marker – assisted selection and Sex manipulation

Molecular or DNA markers are widely used to measure genetic diversity among and between populations (Toro et al., 1999). They are widely used to assess variation in genetic variation (Nielsen, 1998). The use of molecular studies according to Hayes et al. (2006) have been a most successful taxonomic scale, either among population or for the interpretation of intraspecific biogeographic structure. The use of genetic markers to identify loci that control quantitative traits to develop superior strains through marker assisted selection is still in the early stages (Kocher, 1997). Theoretically marker-assisted selection takes a shorter period to improve performance of individuals in a population as compared to conventional breeding. The only constraint to the use of this technique is the cost involved in developing linkage maps (Changadeya et al., 2003).

Sex manipulation in fish involved the methods of gynogenesis and androgenesis. Gynogenesis according to Thorgaard (1986) is an “all maternal” type of inheritance, where the genetic material of the sperm cell does not

contribute to that of the embryo. While androgenesis is the opposite of gynogenesis, all chromosomes of a true androgenetic embryo are of paternal origin. Production of androgenetic diploid individuals is based on inactivation of the genetic materials of the egg, fertilization by normal sperm and blocking of the first mitotic division of the male-origin haploid zygote (Gruinina *et al.*, 1994). Androgenesis has more advantages when compared with gynogenesis for genetic and breeding purposes in fish. This include shorter generation time of males, possibility of sperm preservation and examination of mitochondrial inheritance (Thorgaard, 1986).

The most recent use of this procedure in fish is the production of “YY” male technology. Through a sustained breeding programme the University of Wales, Swansea (UK) and University of Stellenbosch (South Africa) have developed an innovative and robust new genetic technology for producing all male progeny in the Nile tilapia *O. niloticus* known as ‘YY’ male technology (Brink *et al.*, 2002). This is a form of breeding programme combining feminization and progeny testing to produce novel males with YY genotypes (i.e. 2 male sex chromosomes) instead of the usual XY male genotype. Reports from the field across Africa, suggests that gynogenesis increased the yield in many culture systems thereby boosting food production from aquaculture (Jamu and Brummett, 2002).

5. Transgenic

Transgenics in fish is a process whereby a recombinant DNA (transgene) is artificially introduced and integrated into its genome (Palmiter and Brinster, 1985). The expression of the introduced DNA improves one of the aspect of production and both the DNA and the effect are passed to offspring (Fletcher and Davies, 1991). This also involves manipulation of genetic codes to produce transgenic organisms. Since the first transgenic animals were successful with the additional growth gene hormones producing gigantic mice (Palmiter *et al.*, 1982). There is optimism for the application of this aquaculture. In tilapia transgenics that contain the exogenous growth hormone (GH) gene construct demand from chinook salmon have demonstrated growth enhancement (Rahman and Maclean, 1997). According to FAO (1997), transgenic tilapia grows three times more than non-transgene siblings. This also has been reported in common carp, cat fish and coho salmon. This can be achieved in fish by micro injection of the solution of the recombinant DNA into the cytoplasm of the fertilized fish (Lin *et al.*, 1994).

Constraints to the application of fish genetics in aquaculture practices

Utilization and application of genetics in improving aquaculture practice in Nigeria has been limited by.

Limited Research Materials and Facilities and Funding

Research facilities and materials used in fish genetics are scarce in the country and where they are available they are obsolete. Such can not meet the present day challenges in modern technique of aquaculture. In most research activities in fisheries is on going facilities and equipment used in fish genetics are the least thereby putting a constraint to the research works in this crucial aspect of fish management. Insufficient funding is one of the major factors limiting the spread and modernization of aquaculture in developing countries, including Nigeria. This has impeded the growth of aquaculture development in the areas of new techniques and improved technologies needed for the expansion of fish farming to boost fish production. Even in some cases, where there are funds for research in aquaculture only limited amounts are allocated to research in genetics (Yapi – Gnaore, 2002).

Lack of Skills, Technical Know how, and awareness

Lack of adequate skills and expertise training and man power development is one of the factors hindering the application of fish genetics in Nigeria. While there capacity and willingness to conduct research in aquaculture genetics in some institutions in the country, there are few experts with indepth knowledge in this innovative aspects of aquaculture. In fact there is hardly any institutions in the country offering courses in fish genetics. Besides, there is little or no awareness about the potential of genetics in boosting fish production. In most parts of the country the farmers who are the end users of the research findings in this area, are not even aware of this new innovation in fish farming thereby limiting its application in the country.

Inadequate Aquaculture extension services and Hostility from the Public

There is inadequate aquaculture extension service in the country. In some cases where there are major breakthroughs in fish genetics as regards aquaculture, there is little or no channel where this information can be

disseminated to the end users which are fish farmers, thereby limiting the application of fish genetics among fish farmers in Nigeria.

Many people in the country are hostile towards genetically improved fish and products. They have the erroneous belief that all the genetically improved fish are carcinogenic and hence avoid them. This has contributed in no small measure towards limitation of genetic application across the country.

STRATEGIES FOR PROMOTING UTILIZATION OF GENETICS IN AQUACULTURE.

Provision of modern research materials and facilities and Adequate Funding

Modern equipment and facilities should be provided by the government to relevant institution involved in research activities in fish genetics so as to facilitate research in this area. The provision of these equipment will undoubtedly contribute in a greater way to the overall sources of aquaculture in the country, thereby improving the livelihood of the populace.

More fund should be provided for the purpose of research in fish genetics.

Collaboration with Foreign Partners

There should be collaboration with some foreign agencies, for the purpose of technology development and transfer. This is a dynamic process that involves creating, mobilizing, utilizing, enhancing, upgrading and adjusting the existing capacities of individuals and local communities and institutions (Ngude and Sunday 2002). Since technology transfer is a continuous process, the networking with research institutions outside the continent has to be continuous so that Nigeria will not be left behind in fish genetics technology.

Training and Manpower Development and public enlightenment

Scientists from Nigeria should be trained abroad in specific technologies in countries that have made progress in the area of fish genetics. Also training can be received through networking and linkages with relevant institutions across the globe.

Awareness about the potential of genetics in increasing and enhancing production of fish in aquaculture should be disseminated to the fish farmers across the country. Besides, farmers should be supplied and encouraged to grow genetically improved species. Also the general public should be enlightened about the safety of consuming genetically modified fish product.

Conclusions

Genetic improvement of culture species in aquaculture has a vital role to play in boosting fish production in Nigeria. With the constraint identified and the possible solutions enumerated, the way forward therefore is promotion of sustainable genetic improvement programmes that will not compromise, conservation of biological diversity in cultured environment and in the wild. This is very important for the conservation of these species. Also transfer or importation of genetically improved strains from other regions should be controlled. Simple selective breeding of our indigenous species will produce improvements without earning any genetic differentiation to the wild populations.

Application of these techniques together with the local traditional methods will hopefully contribute immensely to the development of a more efficient aquaculture in the future. Hence, all stakeholders in the aquaculture industry must galvanize efforts in formulating and implementing genetic manipulation policy in a sustainable manner to ensure environmental protection and at the same time achieve economic benefits.

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